1. About this Open Learning Module

Target Audience and Prerequisites

This Open Learning module is aimed at new recruits who perform the role of a Signalling Technician for either installation or maintenance.

Apart from a basic Secondary education, an ability to understand moderately complex English language and the presentation of information in the form of diagrams, plus a reasonable curiosity and a willingness to learn, the only requirement is that you should have successfully completed study of the associated Open Learning Module "Introduction to Railway Signalling", or have acquired equivalent basic knowledge about the purpose and implementation of railway signalling.

Aim

This module is intended to provide an introduction to the various forms of diagrams and schematic drawings used in railway signalling, the various symbols and methods of labelling employed and the specialised jargon. It covers

- circuit and wiring diagrams
- red and green colour work
- basic circuit contents
- signalling features

What you will require

Apart from this Open Learning manual the only other material you will require is your Network Rail Rule Book, the Track Safety Handbook and the Signalling Installation Handbook (SIH).

Objectives

When you have completed this module, the assessment and been awarded your certificate, you will

- be able to understand and read circuit and wiring diagrams
- be able to understand and read signalling schematic diagrams
- be able to correctly recognise the commonly used signalling symbols

be able to understand and correctly use the jargon commonly used in association with railway signals and points, and the associated equipment.

An Introduction to Jargon

You may have wondered about the word **jargon**. When talking to a friend you often talk in a way that you both understand but which a stranger may find very difficult to understand. For example, you might say “the Reds were terrific on Saturday”, meaning the local football team had done well.

Throughout this module you will find yourself introduced to a variety of jargon used on the railways. This will help you to understand not only training modules but also some of the language which Signallers, signalling engineers, signalling technicians and others use when speaking or writing about signalling equipment and systems. You will find the same jargon is used in the many written rules, specifications and drawings you will come across in the course of your work.

The Glossary of Terms (see Section 9) introduces you to some of the variations in the jargon.

A Safety Warning

This Open Learning module is designed to arouse your curiosity about railway signalling. However, please remember that, until you have undergone proper training and been certified as competent in Personal Track Safety (PTS) and you have been authorised by your supervisor, you are forbidden to go onto any part of the railway that is not open to passengers, unless you are properly supervised and it is an official part of your training.

Study and remember what it says in your Rule Book and the Track Safety Handbook. There is no requirement for you to go onto the railway to successfully complete this course of study.

Of course, your supervisor or appointed Mentor may use their discretion and properly arrange for you to go onto the railway to look at practical examples of the signals, associated equipment and junctions described in this manual and which you have not already had an opportunity to observe in the course of your work and other training.

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2 Circuit and Wiring Diagrams

2.1 Introduction

In order to communicate effectively it is essential that the message is

- understandable
- concise

A message can take a number of formats, from a telephone conversation to a graph. The rules remain the same, in order to communicate effectively the message must be **clear**.

One form of communication which is frequently used for railway signalling is a **plan** or **diagram** . It is important that the symbols and abbreviations which are used on these diagrams and plans are **fixed**. This means that one section or department will be able to understand immediately what another section or department means.

In Britain there are specific symbols and abbreviations which are used on all plans and diagrams. All symbols and abbreviations are specified in British Standard 376 (BS 376) which is the standard used for railway signalling symbols.

Main Types of Diagram

There are three main types of diagram, namely

- circuit diagrams
- signalling plans
- bonding diagrams

Circuit diagrams can also be called

- wiring diagrams
- prints

A circuit diagram shows electrical signalling circuits and the apparatus used within the circuit. Each component is fully described. The circuit diagram also shows the wire connections between the various components.

Signalling Plan

A signalling plan is basically a type of location map. It provides a detailed layout of all signalling equipment which will include

- track circuits
- points
- signals

Signalling features will be examined in greater detail in Section Four.

Track circuits and the use of the running rails to carry the return currents for DC and AC electric traction requires steps to be taken to ensure the electrical continuity of the running rails. This requires particular attention in the vicinity of points. Electrical bonding wires and jumper cables are used and **bonding diagrams** are used to indicate their locations.

Bonding diagrams can also be called bonding plans.

The utilisation of bonding wires and jumper cables in the context of track circuits is dealt with in greater detail in Module 3: Principles of Track Circuit Operation.

This section will examine

- the Signalling Installation Handbook
- common circuit symbols
- nomenclature
- circuit elements
- termination details
- circuit construction

The Signalling Installation Handbook

Although you will be looking at the components of circuit diagrams, it is important to realise that there is additional information available to supplement this learning material.

The Signalling Installation Handbook section 1D 05 provides detailed examples of the symbols and abbreviations used on

- circuit diagrams
- signalling plans
- bonding diagrams

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If you do not know what a symbol or abbreviation represents, or you just want to check that you have made a correct identification, you can refer to the Signalling Installation Handbook which will be available to you as there is a copy at every depot

Ask your supervisor where it is located and look up the relevant section.

In order to answer certain Self-Assessment Questions you will need to refer to the Signalling Installation Handbook

2.2 Common Circuit Symbols

There are a number of symbols which are regularly used on circuit diagrams. It is important that you are able to recognise these symbols.

The most common of these symbols are shown in Table 1. Look carefully at Table 1, take your time and try to remember what each symbol represents.

Common Circuit Symbols

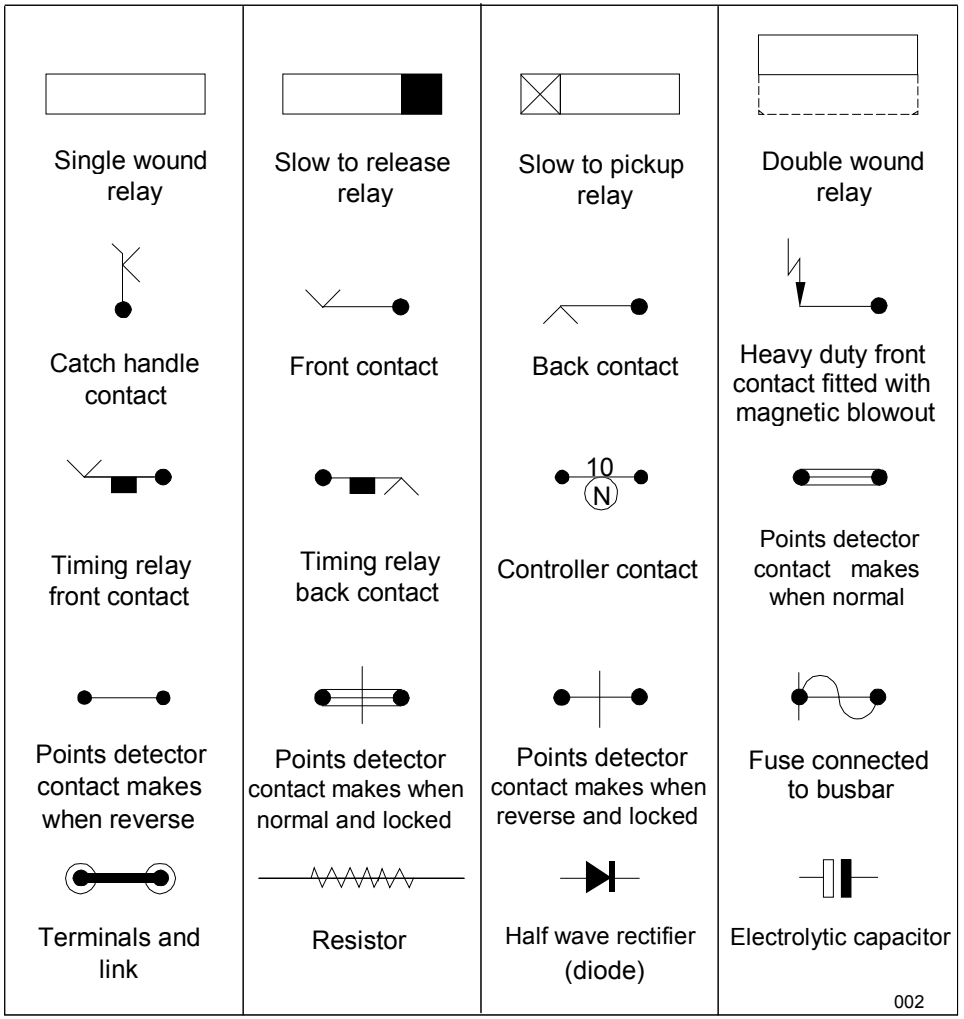


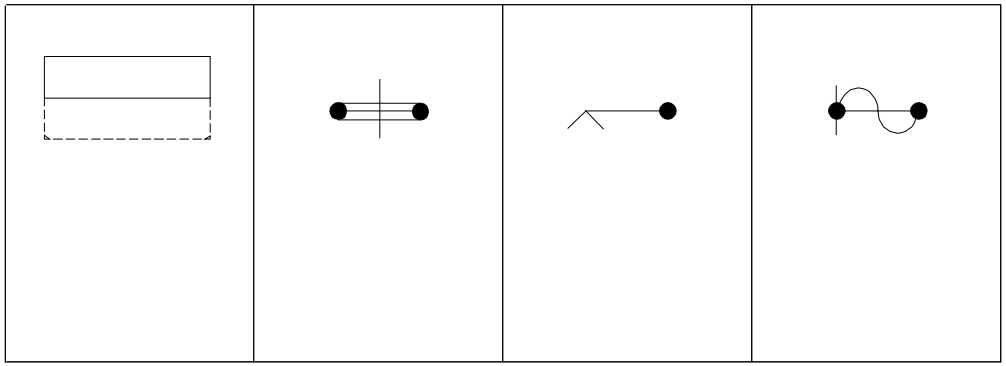
TABLE 1

You can see from Table 1 that each symbol represents a piece of equipment. Certain pieces of equipment may be known by other names, for example, the controller contact is sometimes referred to as a **lever band**. It is not unusual for different terminology to be used in connection with apparatus and practical experience will help you to realise this. However, this learning material will generally use the terms provided in the Signalling Installation Handbook.

Now try to answer the following Self-Assessment Question without referring back to Table 1.

Self-Assessment Question 1

Identify the circuit symbols shown in Table 2.



003

Table 2

Now check your answers with those given in the Answers Section. If you experienced any difficulty, re-read the text until you are confident that you understand the information.

Notation

With all circuit diagrams, two types of notation are used,

- the symbol
- nomenclature

You have already looked at a number of examples of the symbols used in circuit diagrams. You will now examine the second feature in greater detail.

2.3 Nomenclature

The dictionary definition of the term nomenclature is - **a system of naming**. A nomenclature has been produced which is unique to railway signalling. This system incorporates two parts, the

- numerical prefix
- alphabetical term

Numerical Prefix

The numerical prefix is a number used to identify a specific

- lever
- signal
- set of points
- track circuit

Alphabetical Terms

It should be noted that in some cases the prefix for track circuits can be alphabetical.

The **alphabetical term** consists of one or more letters. If a letter is used

- on its own
- after one or more other letters

it represents a **noun a**nd designates the **general kind of unit** .

Preceding letters represent **adjectives** which denote the **purpose of the unit.**

For example, the alphabetical term for a light proving relay is ECR, where

- E indicates light
- C indicates proving
- R indicates relay

The system of nomenclature is described in the Signalling Installation Handbook and in the British Standard BS 376 Parts 1 and 2.

In order to read a circuit it is important that you recognise the abbreviations which are used.

Table 3 lists the abbreviations which are used on circuit diagrams and lists what the abbreviations represent.

Nomenclature (Signalling Abbreviations)

Description (Prefixletter)	Apparatus (Lastletter)
A Approach. Automatic	A
B Block.Bolt	B Block instrument
C Checking. Proving	C Contact
D Clear. Green	D
E Light. Lamp.Heat.Emergency.Earth	E Electric lamp.Earth
F Fog	F Fogging apparatus
G Signal	G Signal
H Caution.Yellow	H Capacitor
HH Preliminary caution.Double Yellow	
I	I Inductor
J Time	J Rectifier
K Indicating. Detecting	K Indicator (visual)
L Locking. Left	L Lock
M Marker. Magnetic	M Motor
N Normal	N Release.Pushbutton.Handswitch. Key
O Retarder	O Resistor
P Repeating	P Lever latch or trigger contact
Q Treadle	Q Local coil of double element relay
R Reverse. Right. Danger.Red	R Relay.Contactor
S Stick	S
T Track circuit	T Transformer.Transmitter
U Route	U Route indicator
V Trainstop	V Trainstop
W Points	W Point machine
X Audible indicator (level crossing)	X Audible indicator
Y Slotting. Disengaging	Y Disengaging apparatus
Z Special	Z Special apparatus 004

Table 3

Combining Letters

The letters shown on Table 3 can be combined to represent **specific components** or **conditions** in a number of areas. You may have noticed that certain letters have more than one description, again practical experience will help you to recognise the correct description of a particular piece of equipment. You will now look at a few of the more common letter combinations used in specific areas, such as

- track circuits
- signals
- points
- power supply labels

Look at these letter combinations carefully and try to remember what they represent. It may help to refer back to Table 3 to see how the combined abbreviations have been constructed within Table 4.

Common Letter Combinations

Track Circuits	
TR	Track Relay
TPR	Track Repeating Relay
T4PR	Track Repeating Relay (fourth in a series)
TSR	Track Stick Relay
Signals	
GR	Signal Relay
HR	Yellow Relay
HHR	Double Yellow Relay
DR	Green Relay
ECR	Light/Lamp Proving Relay
(OFF)GCPR	Signal "OFF" Proving Repeat Relay
(ON)GCPR	Signal "ON "Proving Repeat Relay
Points	
NWR	Normal Points Relay
RWR	Reverse Points Relay
NWKR	Normal Points Indicating Relay
RWKR	Reverse Points Indicating Relay
Labelling of Power	
B	Positive feed of a DCsupply
N	Negative feed of a DCsupply
BX	Feedside of an AC supply
NX	Returnside of an AC supply
Miscellaneous	
BCR	Block Proving Relay
BK	Block Indicator
T/J	Transformer/Rectifier

005

Table 4

Non Standard Abbreviations

You will see that the **labelling of power supplies** does not follow the conventions in Table 3. This subject will be examined in greater detail later in this section as there are specific conventions which apply to this area.

As a general rule any letter that denotes something different to those shown in Table 3 will be either

- in (brackets)
- in “quotes”

Now try to answer Self-Assessment Question 2. If necessary refer back to Table 3.

Self-Assessment Question 2

Identify the following abbreviations shown in Table 5.

NWKE	
WM	
WJR	
EK	
RWKPR	
USR	
ZR	
	006

Table 5

Now check your answers with those given in the Answers Section. You may have wondered which description to use for EK. It was stated earlier that practical experience will help you to recognise the correct description. Do not expect to be able to immediately identify specific types of equipment from the abbreviation alone. You can always refer to the Signalling Installation Handbook or ask your Mentor.

Make sure that you have located the Signalling Installation Handbook before you attempt Self-Assessment Question 3.

Self-Assessment Question 3

On Table 6 use the Signalling Installation Handbook to identify the following symbols and abbreviations.

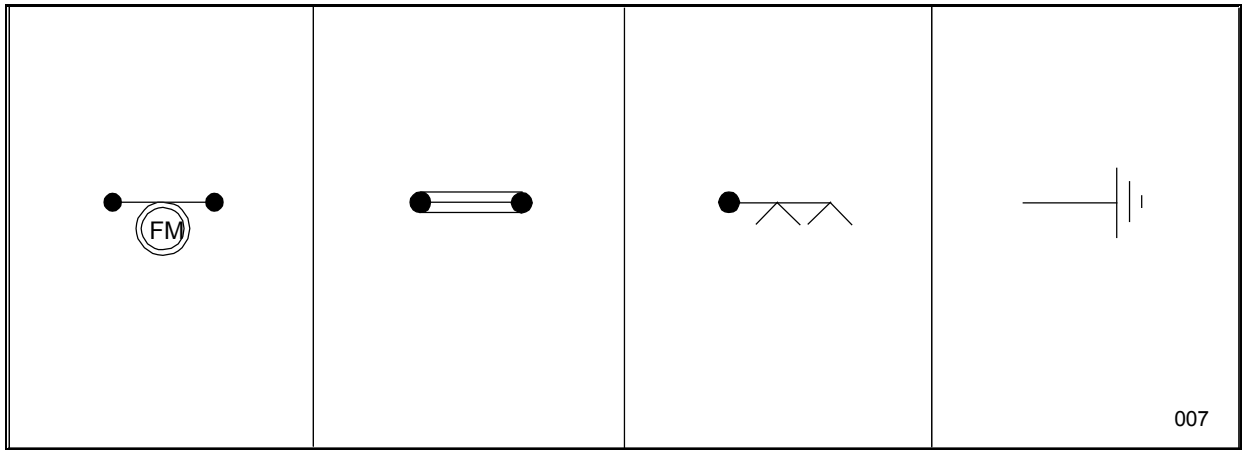


Table 6

Now check your answers with those given in the Answers Section. If you experienced any difficulty, re-read the text until you are confident that you understand the information.

2.4 Circuit Elements

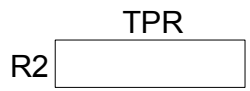
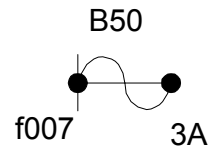
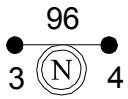
You have now looked at the symbols and abbreviations which are used in circuit diagrams. When the symbol representing a piece of equipment is placed in a circuit diagram it is known as a Circuit Element, as it is an element within a circuit. The circuit element will be labelled with the abbreviations you have already examined.

The abbreviations will provide the following information

- the type of equipment
- a unique identity for this piece of equipment
- termination details

You have already looked at the symbols for equipment such as fuses and relays. On a specific circuit diagram the symbol would include information similar to that shown on Table 7.

Descriptive Examples of Circuit Elements

		
The information on the relay tells us TPR is a track repeating relay of AE track circuit R2 and R4 show where the circuit connections to the coil should be made	B50 -represents a positive dc supply at 50 volts f007- is the fuse number 3A - tells the fuse rating in amperes Note the ampere rating may also be shown as α	Console switch makes when normal 96 is the numerical prefix representing the equipment to be controlled 3 and 4 show where the circuit connections to the switch should be made

008

Table 7

Look carefully how the relay has been labelled in Table 7. There are a number of ways in which **relays** can be labelled. On certain circuit diagrams the information may be presented differently, for example R2 and R4 could be placed **inside** the relay symbol.

By combining the symbol and abbreviation you are now able to clearly identify the specific equipment which is used in the circuit.

You are now going to look at the termination details which will also be shown on the circuit diagram.

2.5 Termination Details

As with the relay shown in Table 7 all pieces of

- equipment
- cables
- wiring

will have **individual termination details** stating where the terminations should be made.

Supply Terminals

Examples of **apparatus** letter codes for **supply terminals** include

Letter	Apparatus
f	fuse
t	terminal

The **description** letter will indicate how the termination is used, examples of these include

Letter	Description
B	Positive feed of a DC supply
N	Negative feed of a DC supply
BN	Intermediate connection to a DC supply
BX	Feed side of an AC supply
NX	Return side of an AC supply
BNX	Intermediate Connection to an AC Supply
E	Earth connection

As with the nomenclature used with apparatus, the information concerning the termination is built up by a combination of information

- numbers added after the **description** letter indicate the voltage
- numbers after the **apparatus** letter indicate the number of the item within a group of items

If you see any termination details on a circuit which you do not recognise, look them up in the Signalling Installation Handbook.

By this stage you have looked at

- circuit symbols
- nomenclature
- circuit elements
- You are now going to consider how this information is constructed into a circuit diagram.

2.6 Circuit Construction

Look carefully at Figure 1 which is an example of a **non-branching path.** This type of circuit is known as a **series circuit.**

Series Circuit

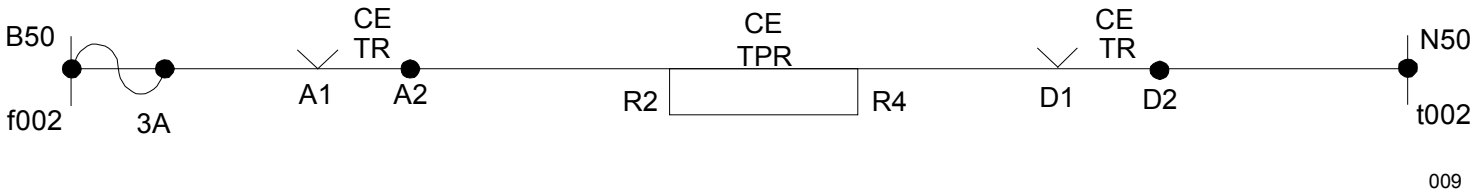


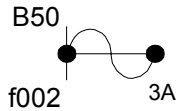
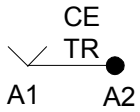
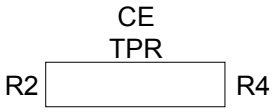
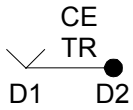
Figure 1

Description of Circuit Elements Figure 1

You should be able to identify the element functions in this diagram by either

- memory
- looking up the unfamiliar symbols and abbreviations in the Signalling Installation Handbook

A full description of Figure 1 is shown in Table 8.

	B50 represents the positive dc supply at 50 volts f002 is the fuse number 3A is the fuse rating of 3 amperes
	This represents a relay front contact ∨ represents the contact ● represents the arm CE is the track circuit reference. TR is the track relay. A1 and A2 show where the connections to the relay contacts should be made.
	This represents a track repeating relay CE is the track circuit reference TPR identifies it as a track repeating relay R2 and R4 are the relay coil connections
	This also represents a relay front contact ofthe CE TR. It is said to be “doublecutting” the relay coil.
	N50 is the negative dc supply at 50 volts t002 is the negative termination number The short vertical line through the fuse and negative terminal symbols indicate they are connected to a busbar

010

Table 8

Parallel Paths

The wiring diagram will also show the different paths which the current can take. Look carefully at Figure 2

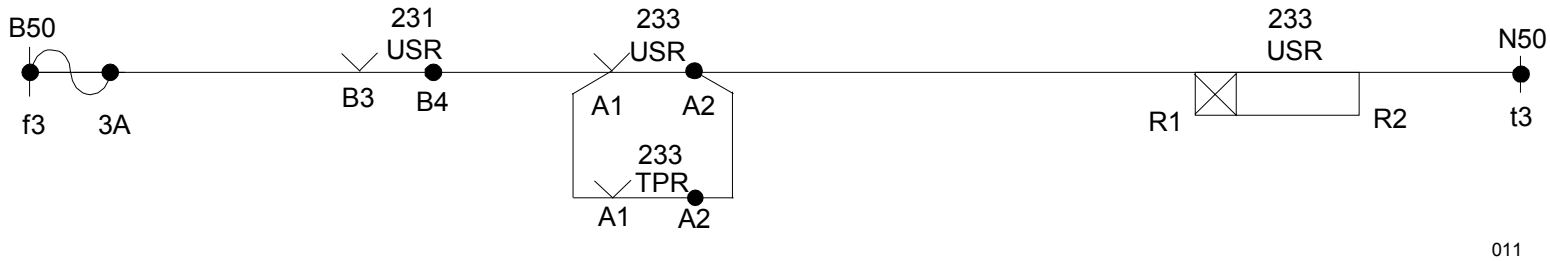


Figure 2

In Figure 2 look at the two front contacts

233 USR

233 TPR

These contacts are positioned one above the other and are connected in **parallel.**

Now look at Figure 3

Branching Paths

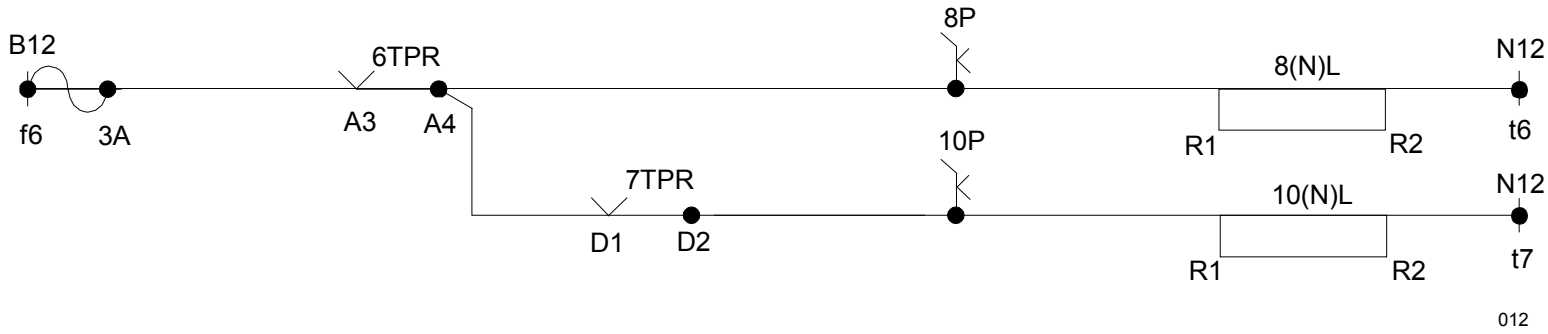


Figure 3

In Figure 3 the A4 connection of 6TPR has **two** wires. At this point the circuit diagram diverges into two separate paths. This diversion is known as a **branching path**.

The circuits which you have been looking at are **simplified.** A typical circuit diagram will usually contain far more information. However, providing you remember to look up in the Signalling Installation Handbook any symbols or abbreviations which you do not recognise, you will be able to understand the basic components of any circuit or wiring diagram.

Self-Assessment Question 4

Identify the component parts shown on Figure 4.

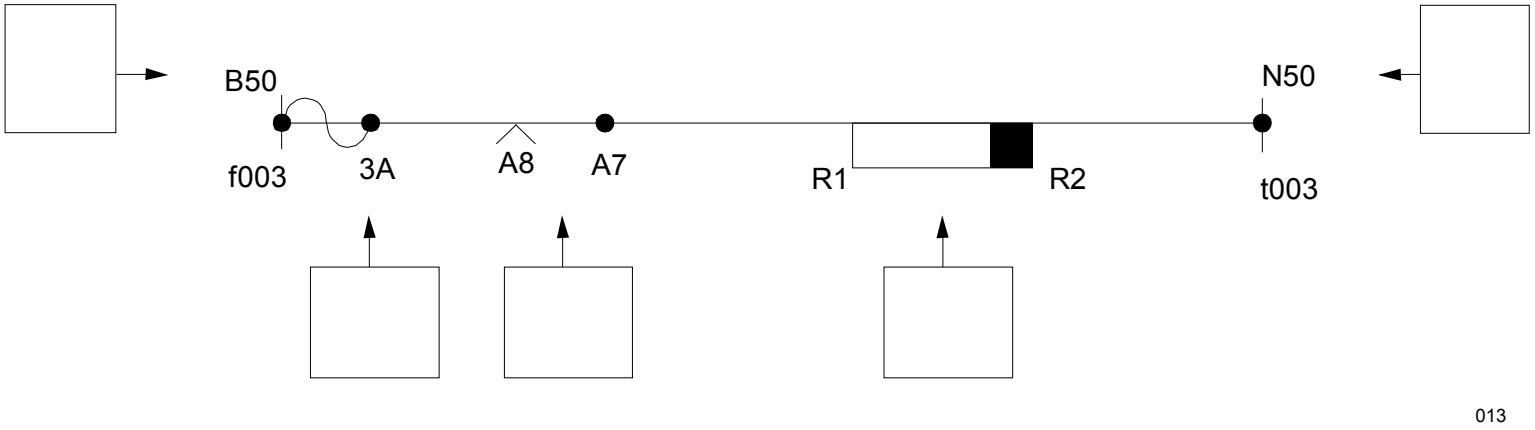


Figure 4

Now check your answers with those given in the Answers Section . If you experienced any difficulty, re-read the text until you are confident that you understand the information.

2.7 Section Summary

This section has looked at

- circuit diagrams
- signalling plans
- bonding diagrams
- the Signalling Installation Handbook
- common circuit symbols
- nomenclature
- circuit elements
- termination details
- circuit construction

Before moving on to Section Three make sure that you fully understand all the information given in Section Two. If there are any areas which you do not understand, re-read the relevant parts of the text.

3 Red and Green Colour Work

3.1 Introduction

Circuits will require

- installation
- possible modification

If either of these operations is to be carried out, it is necessary to have some method of identifying on circuit diagrams what is

- being added
- being removed or recovered

Circuit diagrams are usually printed in black on a white background and it would be complicated if installation or modification work was not clearly identified. A method is necessary which highlights modifications. The system which the railways use involves colour identification.

You will now look at this system in greater detail.

3.2 Red and Green Markings on Circuit and Wiring Diagrams

It is important to remember that

Colour Coding

- new equipment or wiring is marked in red
- the equipment or wiring being removed or recovered is marked in green

A method which you can use to remember this involves the traffic light image.

Method of Renumbering

If you bear in mind that green means go, this image will help you to correctly identify the type of modification taking place.

If the colour code applies to a whole diagram sheet you could see an alternative method. This method would involve a normal black on white diagram which has

Alternative Method of Colour Coding

- an endorsement in red, all new work if it is new
- a green cross through the whole sheet if it is to be recovered

In some circuit diagrams both new and recovered work may be shown on the same sheet. If this is the case the wires which are going to be removed are marked with small green crosses as well as being coloured green.

Complex Alterations

If the alterations are extremely complex and involve large-scale recoveries or modifications, two copies of each drawing will then be issued. These will be

- a first copy showing the wiring and equipment to be removed in **green**
- a second copy showing the new wiring and equipment in **red**

Look carefully at Figure 5

Relay Contact being added to a Circuit

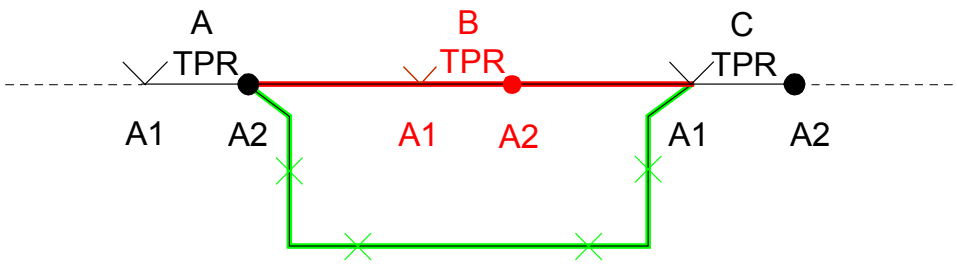


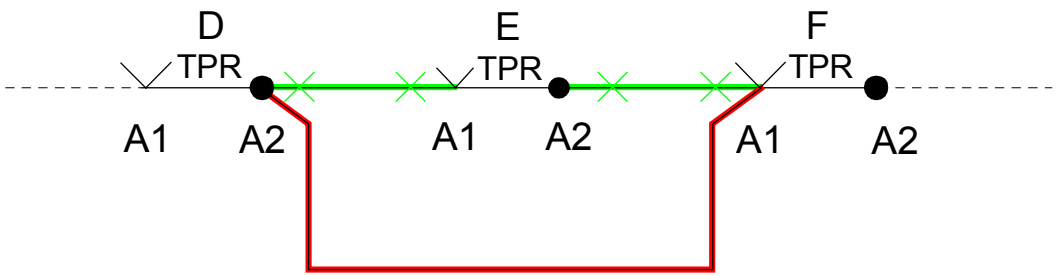
Figure 5

In Figure 5 you can see that a **relay contact** is being added to the circuit. This relay contact and the new wires connected to it are therefore marked in **red**. In this case a wire is removed to make way for the added relay contact. This wire is marked in **green** and is also marked with **green crosses**.

Note that on Figure 5 the **whole** of the relay contact is coloured red. This donates that the relay is new. If a contact of an existing relay is being added to the circuit then only the termination details A1 & A2 would be coloured red.

Relay Contact being made Redundant to a Circuit

Now look at Figure 6



015

Figure 6

Figure 6 shows the contact of an existing relay being made redundant. The **green** lines and crosses indicate which wires are being removed. Note that on figure 6 only the **wires** to the relay contact are being removed. The relay contact, now redundant in this circuit, is not physically being removed, it may be used in another new or existing circuit. This is the reason why the contact is not coloured. The **red** lines indicate what is being added.

This means that a wire will now connect A1 of F TPR to A2 of D TPR . Due to local practice you may see slight variations in use for red and green work.

- green crosses may be black
- a block of red colour may be used instead of an individual red coloured contact, arm, wire and nomenclature

a block of green colour may be used instead of an individual green coloured contact, arm, wire, and nomenclature.

Further Examples of Red and Green Work

3.3

Further Examples

The Signalling Installation Handbook provides examples of red and green work and illustrates how removed elements can be replaced in another circuit. It is worthwhile checking the Signalling Installation Handbook to see how these circuits are marked up.

If this type of work is to be carried out it is important that you **fully understand** the instructions given in the diagram.

Section Summary

This section has looked at

- red colour work
- green colour work

Now complete Self-Assessment Question 5 before proceeding to Section Four.

Self-Assessment Question 5

In your own words state why red markings are used on circuit diagrams

Before moving on to Section Four make sure that you fully understand all the information which was given in Section Three.

If there are any areas which you do not understand re-read the relevant parts of the text.

4 Basic Circuit Contents

4.1 Introduction

You have seen how a circuit diagram is built up by using the symbols to represent equipment and that the symbols are connected by straight lines which represent connecting wires. By using the standardised symbols and nomenclature the person reading the diagram is able to identify

- specific items of equipment
- inter-connections between equipment
- terminals and connecting points on equipment

4.2 Circuit Diagram Title Block and Index Sheets

Typical circuit diagrams run to many sheets. Each diagram has an identifying title block. The title block includes the name and number for identification, version reference and sheet number. The first sheet usually carries an index for the subsequent sheets.

4.3 Identifying Circuit Contents

Look carefully at Figure 7

Discrete Circuits

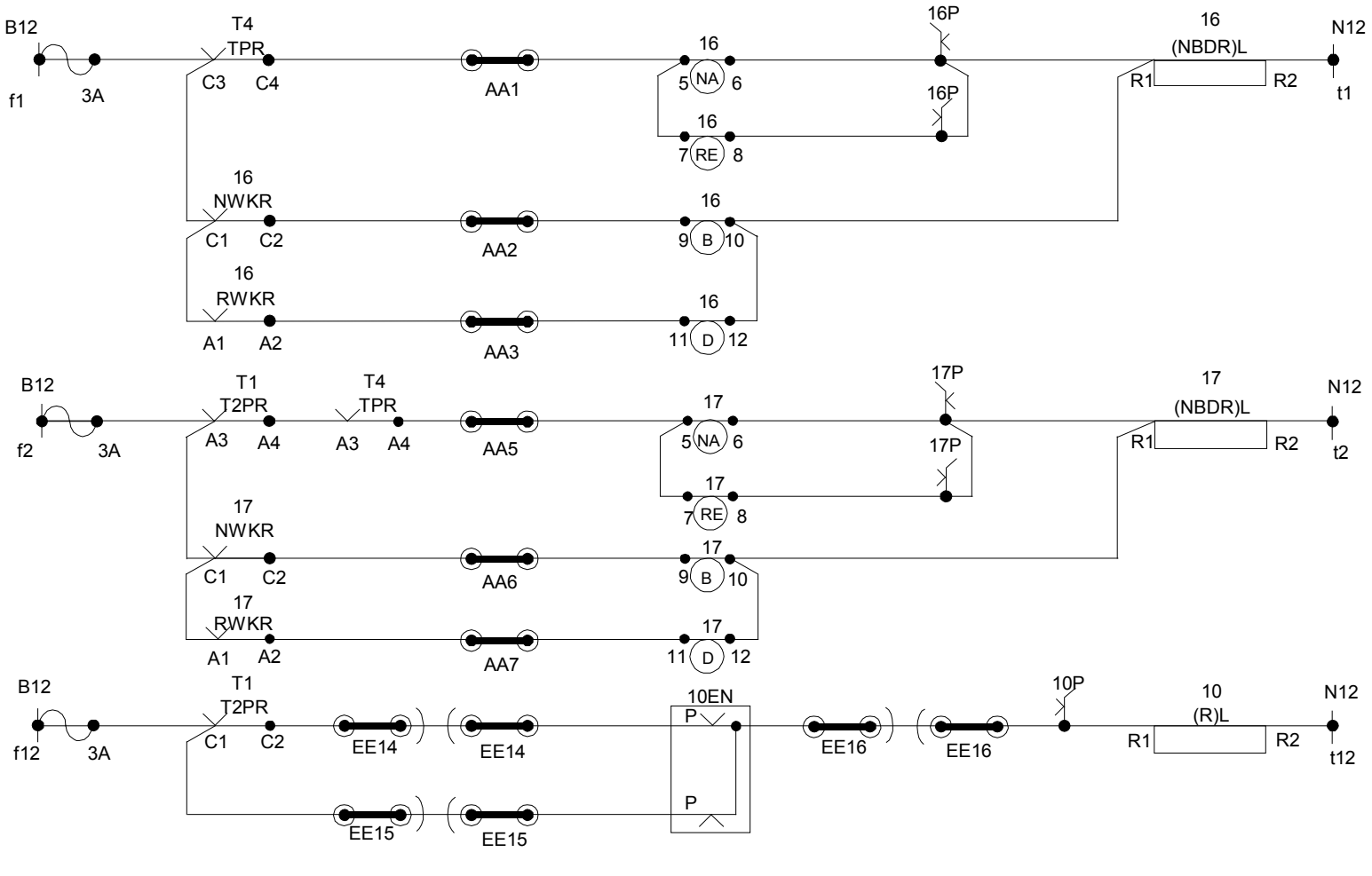


Figure 7

Discrete Circuits

Although Figure 7 is more complicated than the circuits you have previously looked at you should be able to recognise **some** of the components.

Within Figure 7 there are three discrete circuits. A discrete circuit has its own

- incoming power supply, in this case the positive supply
- return power supply, in this case the negative supply

Any sheet of a circuit diagram may contain a number of discrete circuits.

It is possible to

- loop the positive supplies together
- loop the negative supplies together

Now look carefully at Figure 8 which provides an example of positive and negative looping.

Positive and Negative Looping on One Sheet

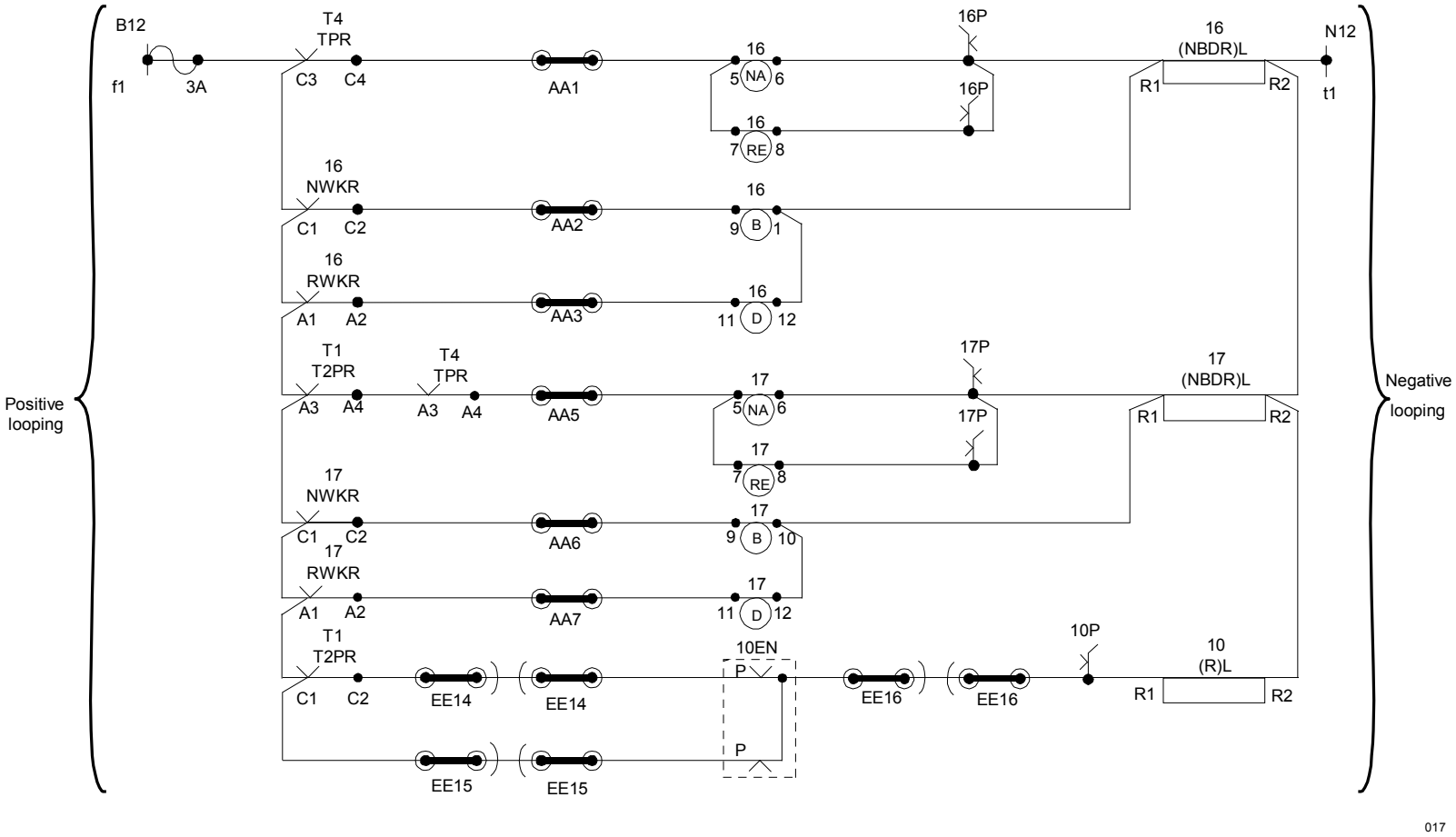


Figure 8

Positive and Negative Looping

In Figure 8 the

- positive supplies have been looped together to form positive looping
- negative supplies have been looped together to form negative looping

It is possible to feed a number of circuits from one source. When this happens the circuit diagram may become cluttered with information. In order to clarify the information, two separate diagrams can be used.

Examples of this can be seen in figures 9 and 10.

In figure 9 the following information is given

- positive and negative and feed supply
- fuse number and rating
- contact number at the start or finish of a circuit being fed
- sheet number

The sheet number represents which sheet of the circuit diagrams will show the full circuit being fed.

Figure 9 would be known as the positive and negative looping sheet.

Positive and Negative Looping Sheet

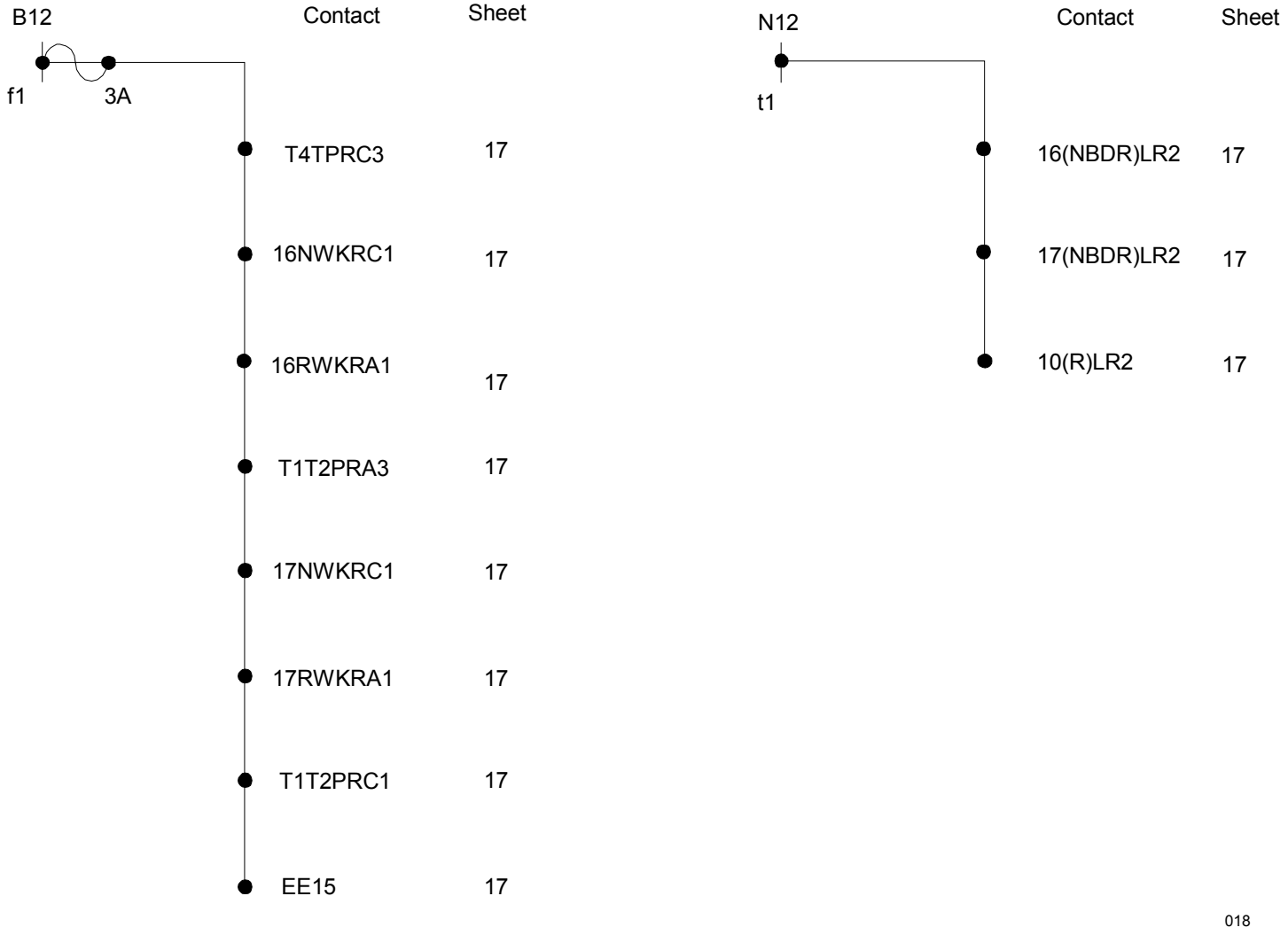


Figure 9

Circuits Without Positive and Negative Looping Shown

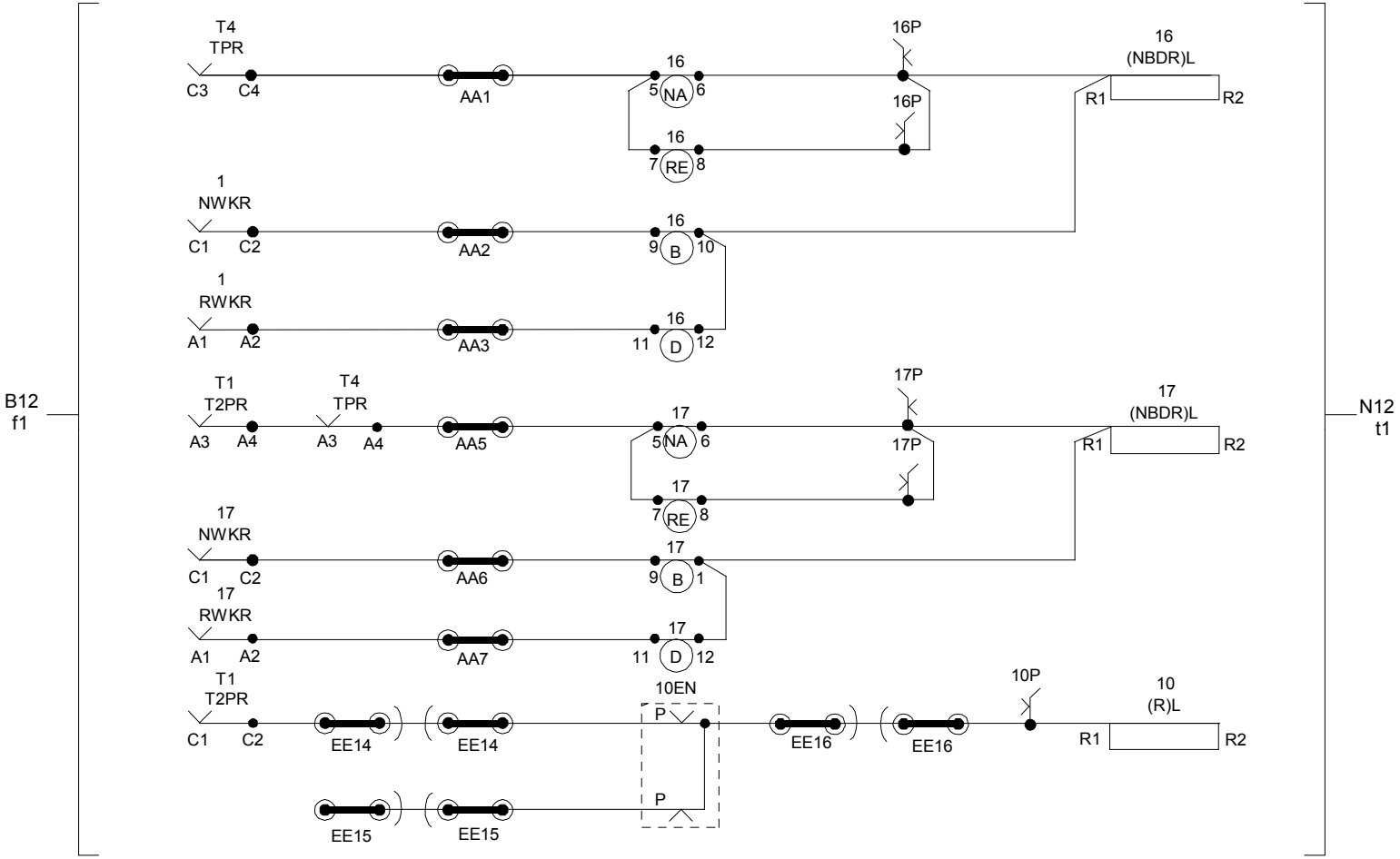


Figure 10

In figure 10 the positive and negative looping is not shown, but it provides detailed information concerning the whole circuit.

4.3 Incoming and Outgoing Paths

Outgoing circuits are feeds from the signal box or relay room to track side equipment.

Incoming circuits are feeds from the track side equipment to the signal box or relay room.

Incoming and outgoing circuits are shown on Figure 11.

Figure 11 shows the outgoing control feed that operates 17 points and the incoming indication feed back from these points.

Outgoing Circuit

Incoming Circuit

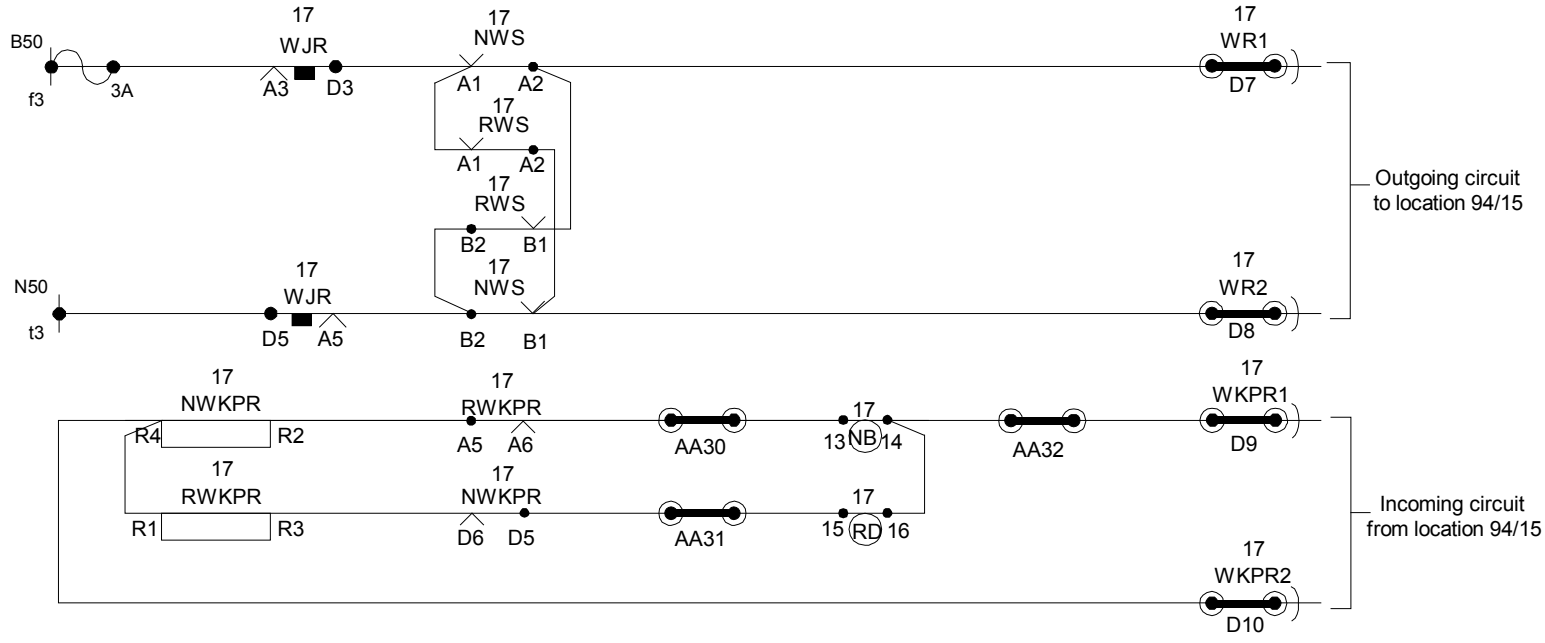


Figure 11

4.4

Interlinking Cables

Interlinking cables are cables used to connect items of equipment within a relay room or signal box.

Figure 12 provides an example of **interlinking cables**.

Interlinking Cables

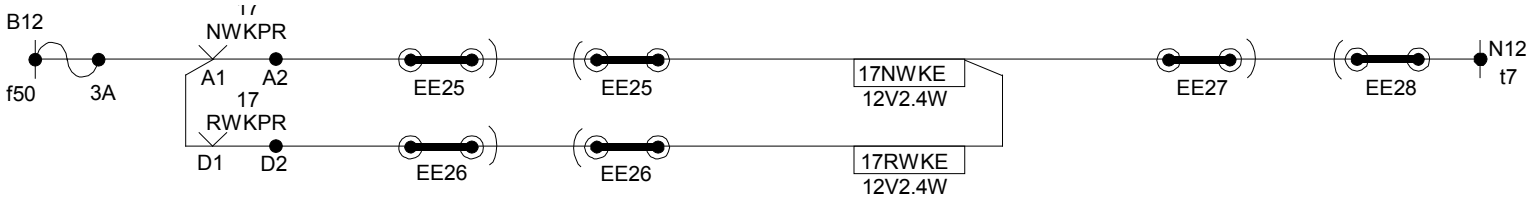


Figure 12

In Figure 12

- NWKPR A1-A2 and RWKPR D1-D2 are the contacts of the normal and reverse point indicating repeat relays which are located in the relay room
- NWKE and RWKE which are the normal and reverse point indicating lights which are located on the block shelf on the operating floor

They are connected by **interlinking cables**.

The interlinking cables are identified by the symbols **)** and **(**

Interlinking cables can also be termed **inter-rack connections**.

Self-Assessment Question 6

Look carefully at Figure 13 and use brackets and arrows to correctly identify the following areas

- positive looping
- negative looping
- interlinking cables

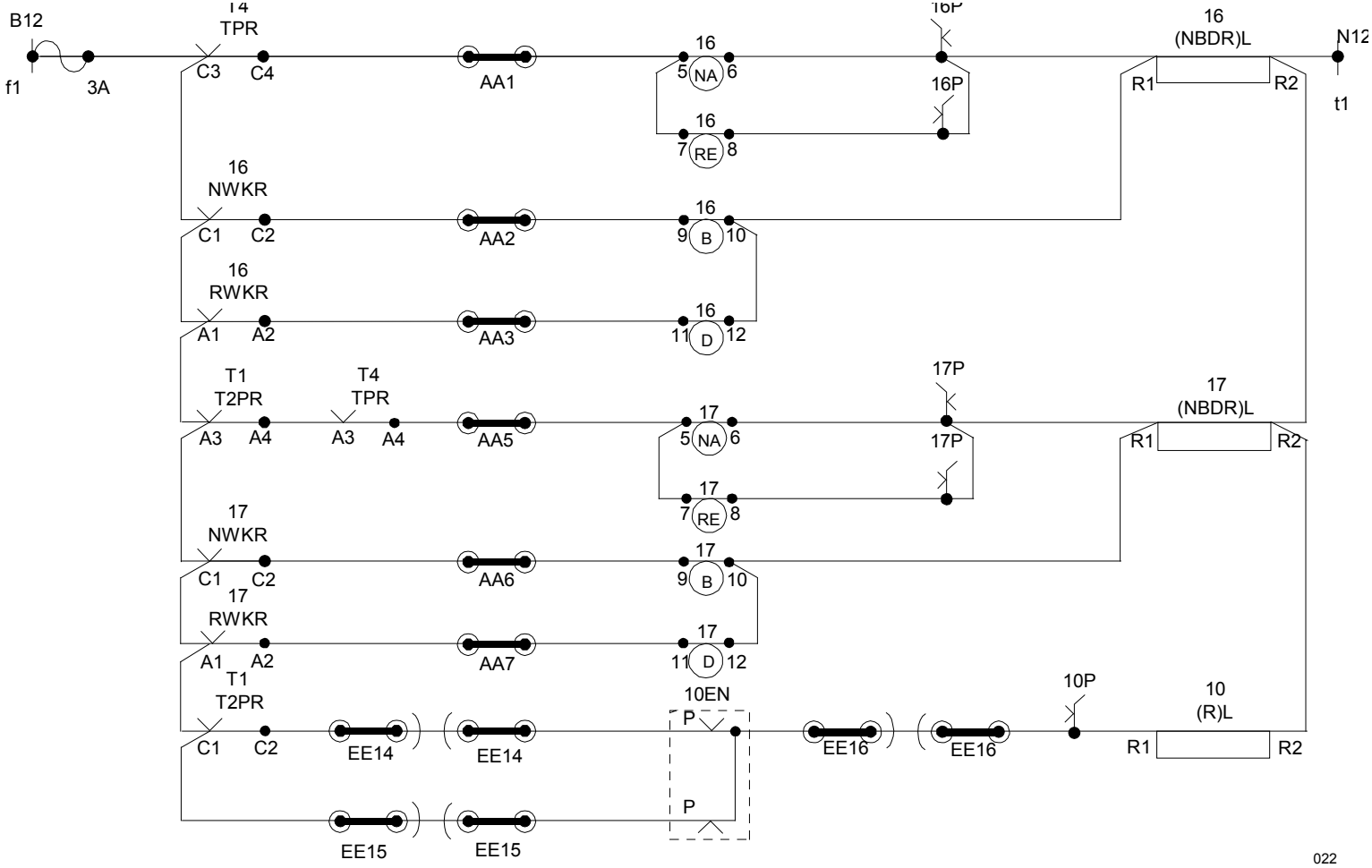


Figure 13

Now check your answer with the one given in the Answers Section . If you experienced any difficulty, re-read the text until you are confident that you understand the information.

4.5 Section Summary

This section has looked at

- discrete circuits
- positive and negative looping
- incoming and outgoing paths
- interlinking cables

Before moving on to Section Five make sure that you fully understand all the information which was given in Section Four. If there are any areas which you do not understand re-read the relevant parts of the text.

5 Signalling Features

5.1 Introduction

A signalling plan is a diagram which represents the actual layout of the signalling equipment including

- signals
- points
- track circuits

You will now examine the different symbols used within a signalling plan.

5.2 Signalling Plan Symbols

Table 9 provides you with some examples of the symbols and abbreviations used on signalling plans.

Look at them carefully and try to familiarise yourself with each one.

Signalling Plan

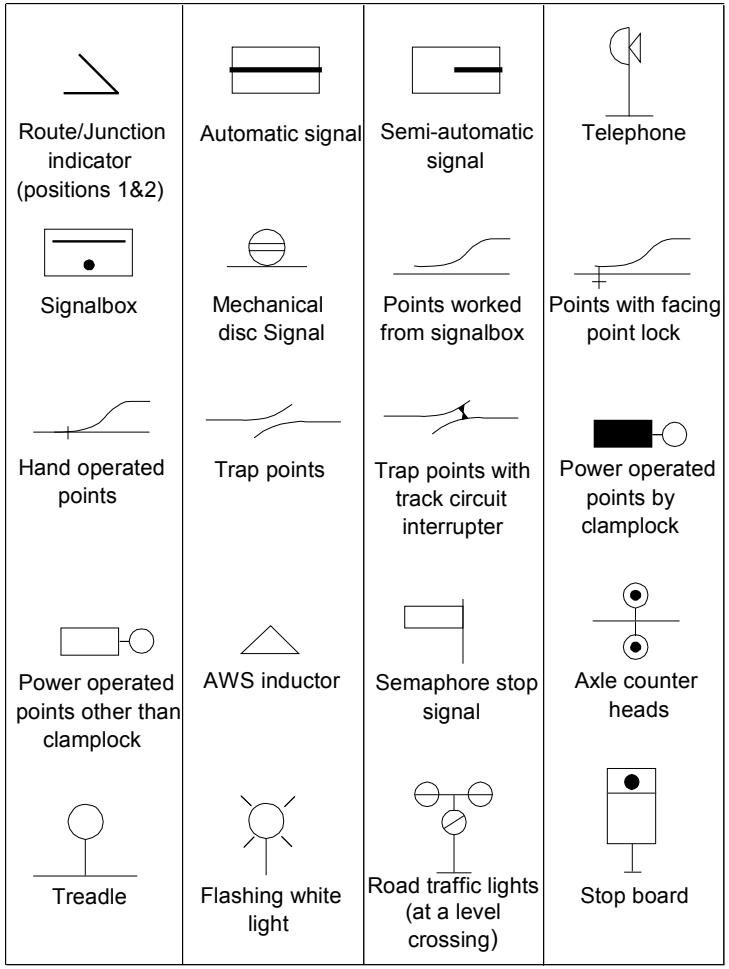


Table 9

The symbols shown on the top line of Table 9 constitute part of the signal make up. An explanation of this can be found in the Signalling Installation Handbook 1D 05 fig. 2.

If symbols or abbreviations are used which you are unfamiliar with, always remember to use the Signalling Installation Handbook to check what they represent.

Now look at Figure 14 which gives you an example of a simplified signalling plan, which shows the section around Toledo Station signal box. Note the gradient and sketch maps which show the **environment** of Toledo Station.

Signalling Plan

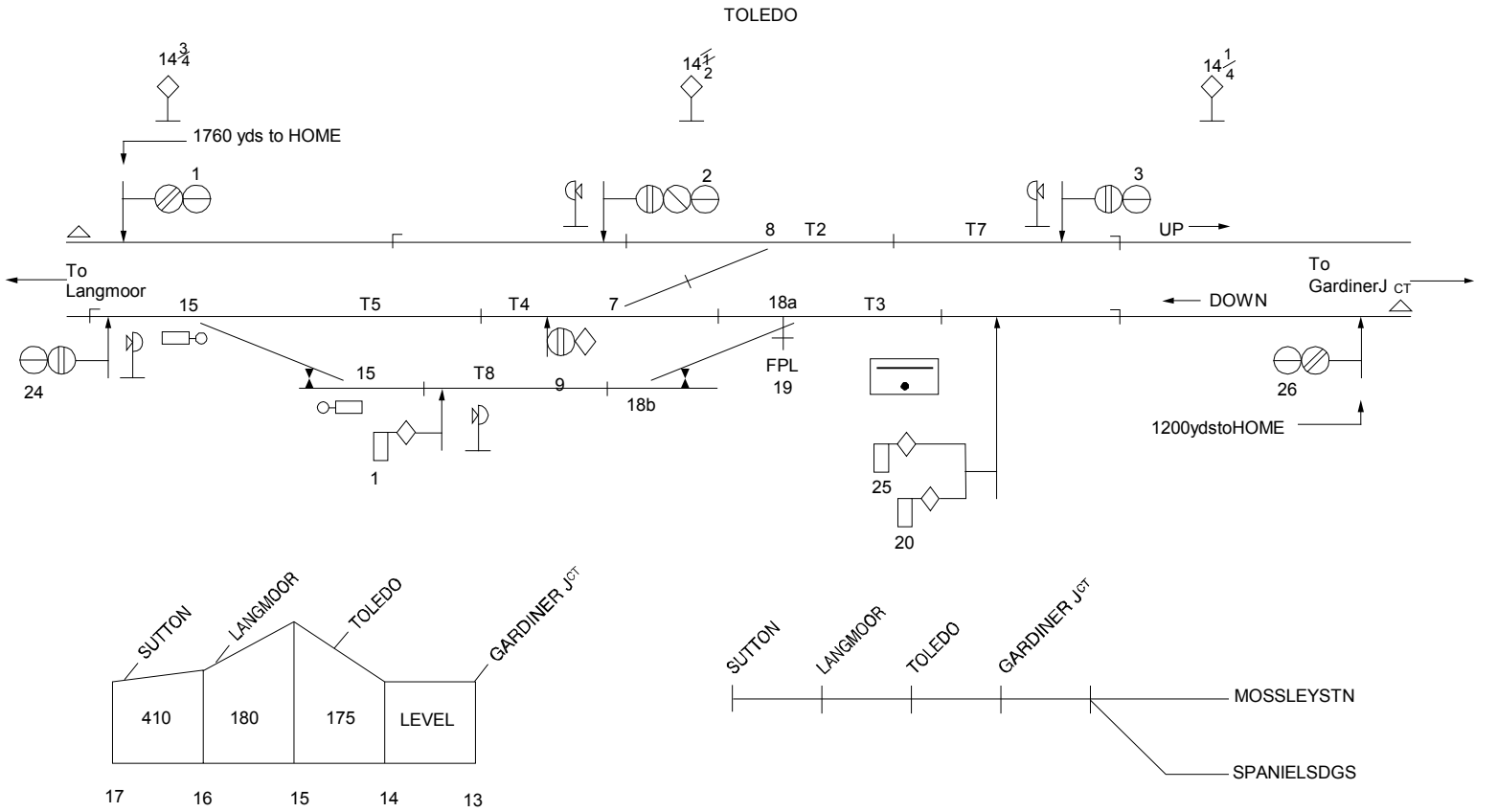
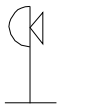
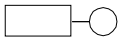


Figure 14

There are symbols on Figure 14 which you should either

- recognise
- look up in the Signalling Installation Handbook

Table 10 identifies some of the symbols shown on Figure 14

	Telephone
	Three aspect colour light signal on straight post
	Power operated points not clamplock
	Signal box
	Mile post

025

Table 10

5.3 Section Summary

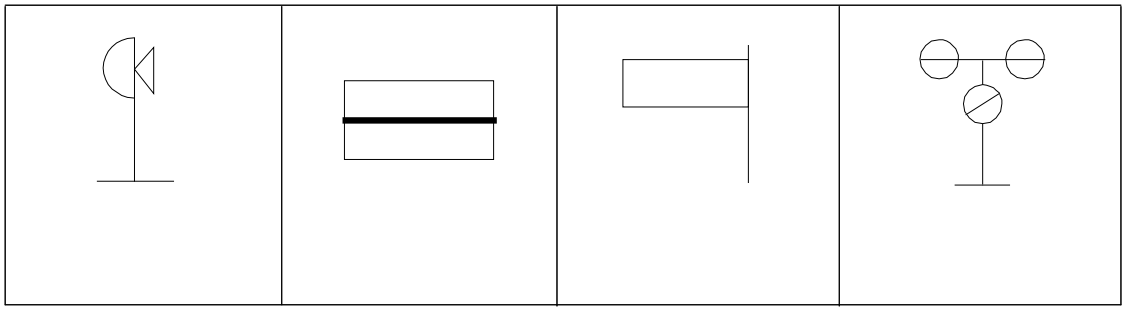
This section has looked at

- signalling plans
- signalling plan symbols

Now complete Self-Assessment Question 7.

Self-Assessment Question 7

On Table 11 identify the following symbols and abbreviations which could be used on a signalling plan.



026

Table 11

Now check your answers with those given in the Answers Section. If you experienced any difficulty, re-read the text until you are confident that you understand the information.

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6 Module Summary

This module has looked at

- circuit diagrams
- red and green work
- signalling features

Before attempting the Self Test Questions which follow make sure that you fully understand what you have read. Take your time and check your answers to the Self-Assessment Questions again.

When you feel confident begin the Self Test Questions

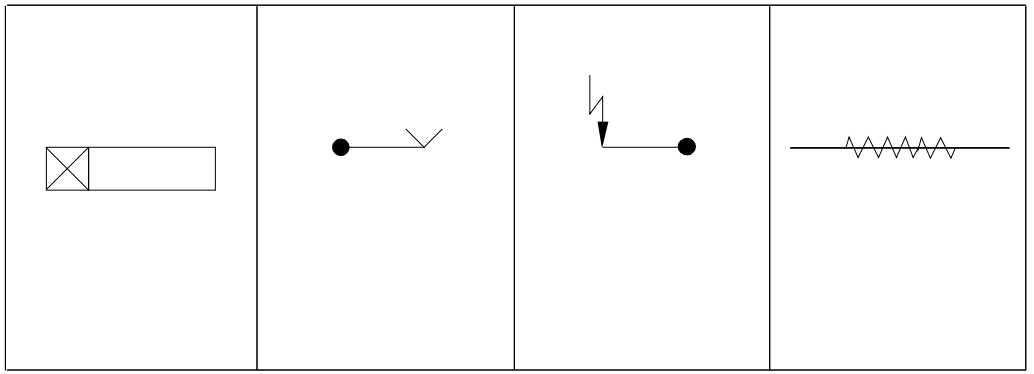
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7 Self Test Questions

Attempt the following Self Test Questions. Take your time, and think about the answers carefully. When you have finished the Self Test check your answers with those given in the Answer Section. If you experienced any difficulties return to the relevant part of the text and re-read the material. Do not move on to the next module until you feel confident in your understanding of the learning material.

Self Test Question 1

Identify the following circuit symbols shown in Table 12.



027

Table 12

Self Test Question 2

State the meaning of the following abbreviations shown in Table 13. If you wish you may refer to Table 3 to help you answer this question.

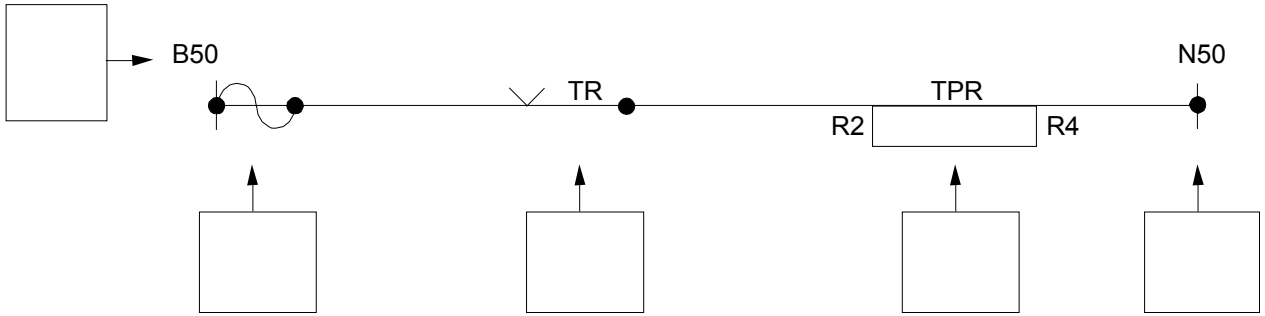
UCR	
BJN	
RLKPR	

028

Table 13

Self Test Question 3

Identify the components on the simplified circuit diagram shown in Figure 15.



029

Figure 15

Self Test Question 4

Look at Figure 16 and explain the alteration to take place.

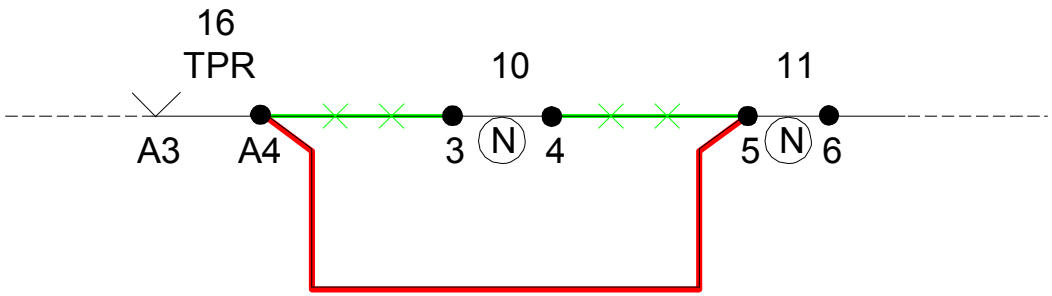


Figure 16

Self Test Question 5

Look carefully at the signalling plan shown in Figure 17 and correctly label the symbols.

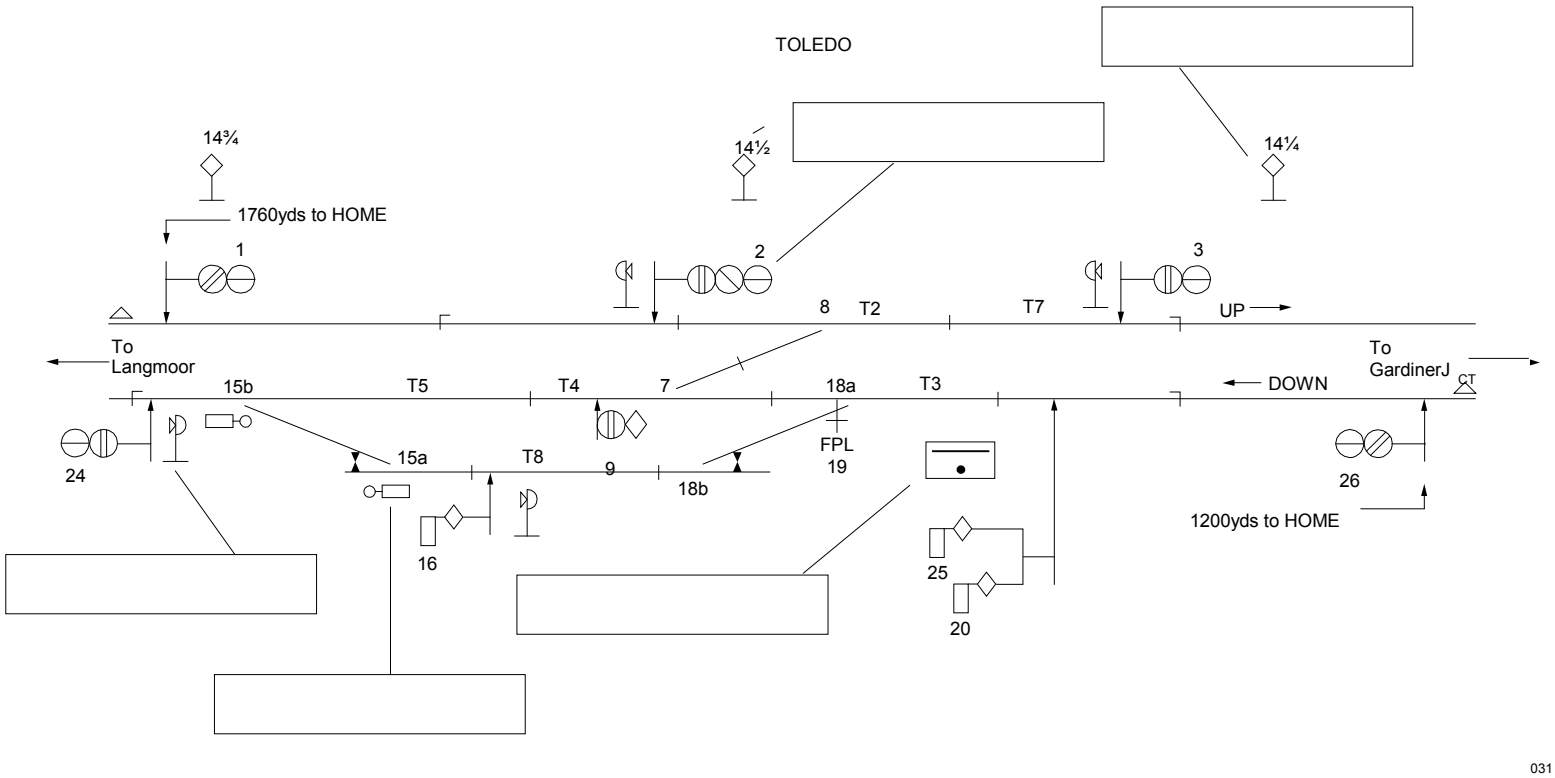

TOLEDO
14¾
14½
14¼
1760yds to HOME
1
2
3
8
T2
T7
UP
To
Langmoor
To
GardinerJ
15b
T5
T4
7
18a
T3
DOWN
24
15a
T8
9
18b
FPL
19
26
1200yds to HOME
16
25
20
031

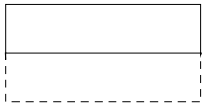
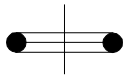
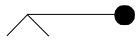
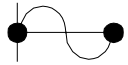
Figure 17

8 Answers

Answers to Self-Assessment Questions

This section provides the answers to the Self-Assessment Questions in Module One. Compare them to your own answers. If you experience any difficulty, re-read the relevant part of the text.

Answer to Self-Assessment Question 1

			
Double wound relay	Points detector contact makes when normal and locked	Back contact	Fuse connected to busbar

032

Table 14

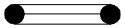
Answer to Self-Assessment Question 2

NWKE	Normal points indicating lamp	
WM	Points motor	
WJR	Points timer relay	
EK	Light indicator	
RWKPR	Reverse points indicating repeating relay	
USR	Route stick	
ZR	Special relay	

033

Table 15

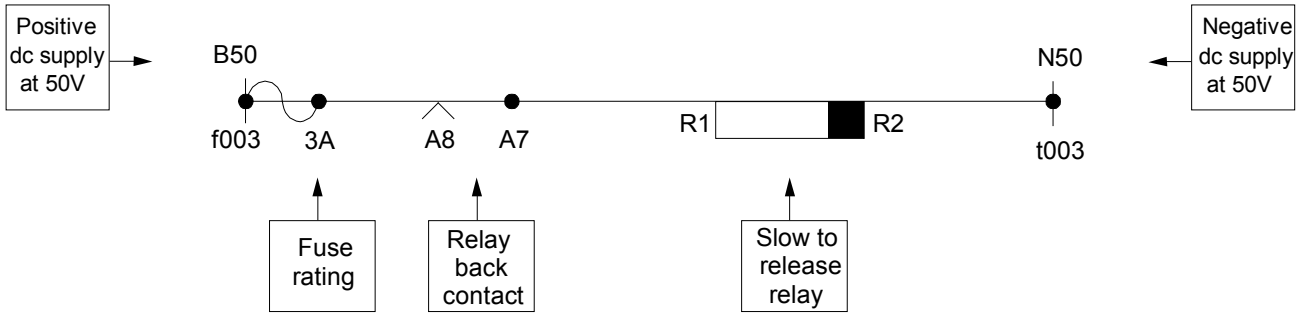
Answer to Self-Assessment Question 3

			
Push Button Contact Breaks when pulled	Points Detector Contact Makes when normal	Lock proving contact	Earth

034

Table 16

Answer to Self-Assessment Question 4



035

Figure 18

Answer to Self-Assessment Question 5

Red markings are used on circuit diagrams to indicate that an **addition** is being made to the circuit.

Answer to Self-Assessment Question 6

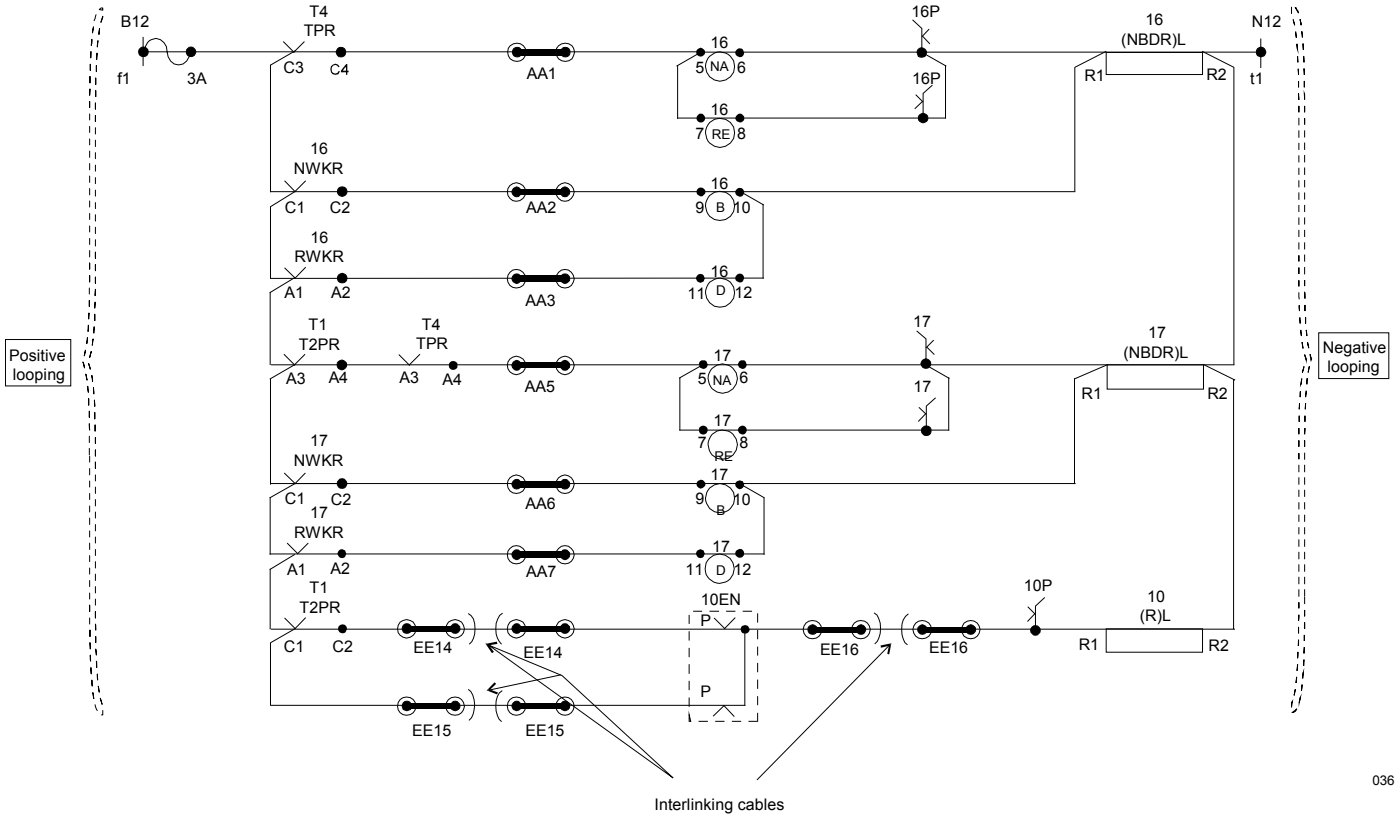
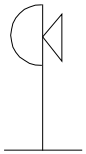
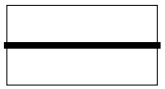
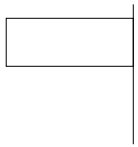
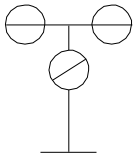


Figure 19

Answer to Self-Assessment Question 7

			
Telephone	Automatic signal	Semaphore stop signal	Road traffic lights (at a level crossing)

037

Table 17

Answers to Self Test Questions

This section provides the answers to the Self Test Questions in Module One. Compare them to your own answers. If you experience any difficulty re-read the relevant part of the text.

Answer to Self-Test Question 1

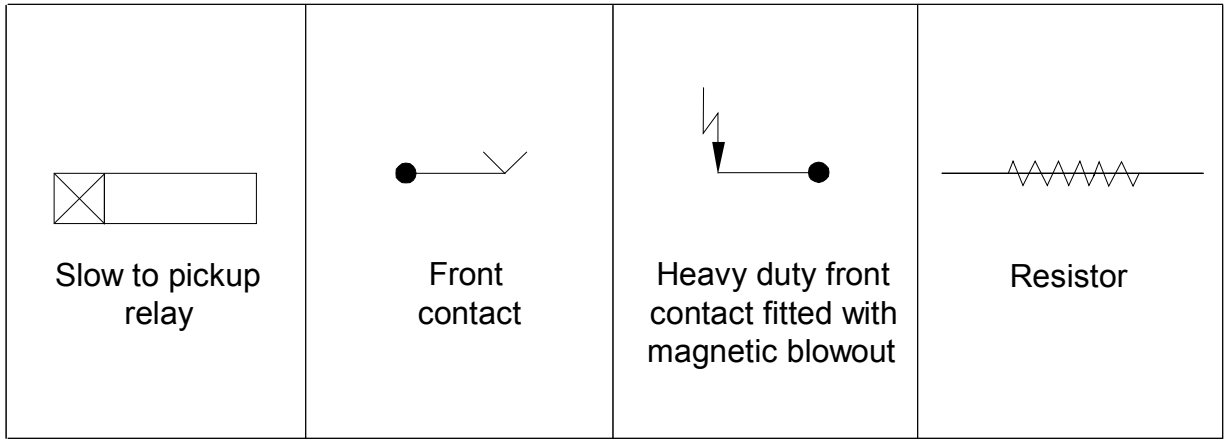


Figure 18

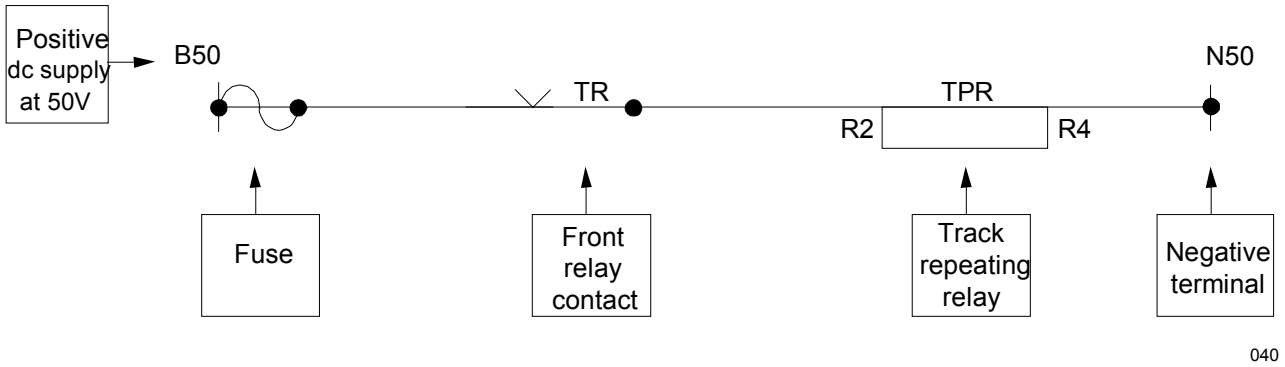
Answer to Self-Test Question 2

UCR	Route checking relay
BJN	Block time release
RLKPR	Reverse locking indicating repeating relay

039

Table 19

Answer to Self-Test Question 3



040

Figure 20

Answer to Self-Test Question 4

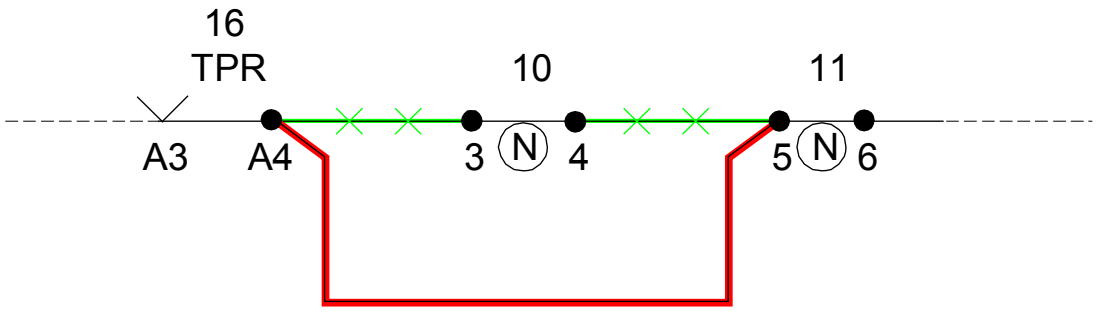


Figure 21

On Figure 21 it can be seen that the

- wires to the controller contact shown in green are being removed from the circuit
- wire shown in red is being added to the circuit, which has the effect of making the controller contact redundant

Answer to Self-Test Question 5

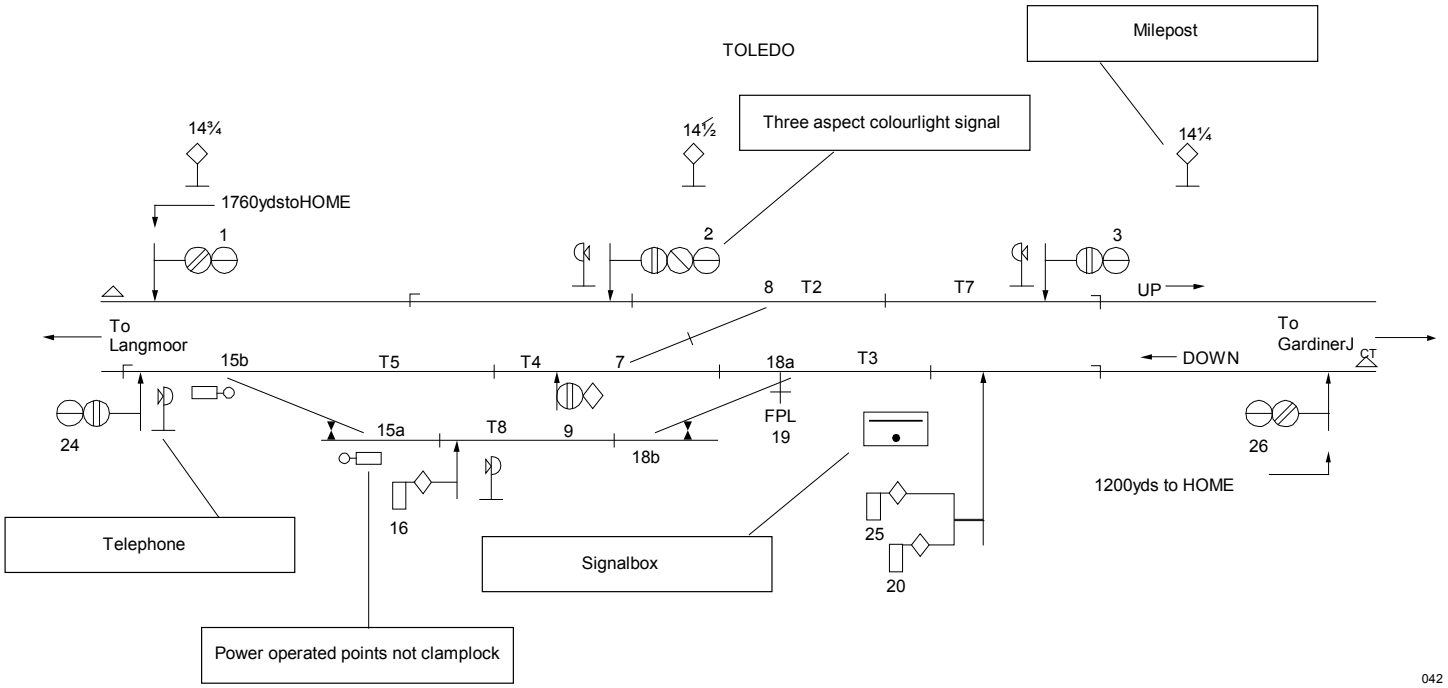


Figure 22

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9

Glossary of Terms

Within the railway system there are a number of regional variations and individual conventions which mean that alternative terms may be used for

- equipment
- operations
- features

Module One has used terms which normally correspond with those found in the Signalling Installation Handbook, however it may be useful for you to familiarise yourself with the alternative terminology which you may encounter in your work. This glossary provides a number of variations which may occur.

Circuit Diagram	wiring diagrams
	drawings
	prints
	comics

Signalling Plan - signalling sketch
scheme plan
domestic diagram

Bonding diagram - bonding plan

Controller contact - lever band
controller band

Interlinking cables - Interlinking connections
inter-rack connections
inter-rack cables

Double cutting - Protects against earth faults